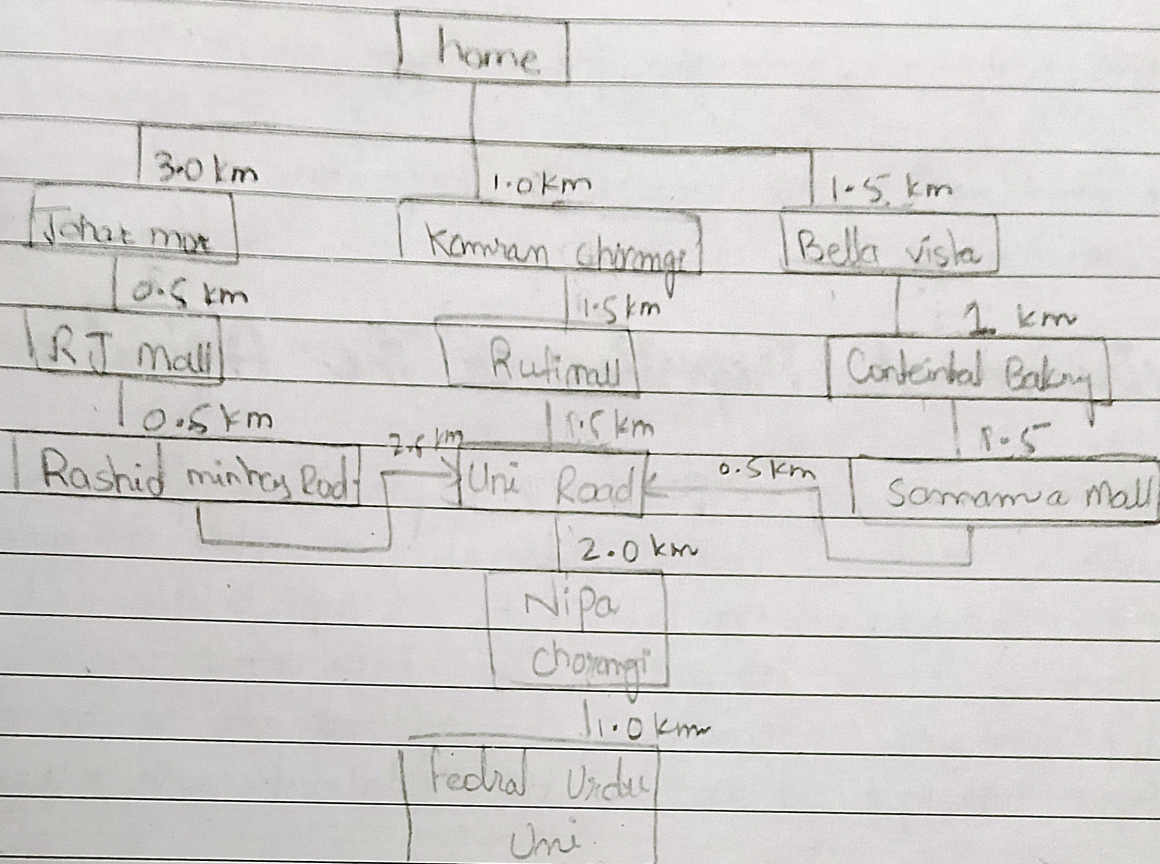


Dated: _____

Home To Uni Route (BFS):



$$R_1 \rightarrow H \rightarrow 3.0 + 0.5 + 0.5 + 2.0 + 2.0 + 1.0 = \text{Total} = 8.5 \text{ km.}$$

$$K_1 \rightarrow H \rightarrow 1.0 + 1.5 + 2.0 + 1.0 = \text{Total} = 7.0 \text{ km.}$$

$$B_1 \rightarrow H \rightarrow 1.5 + 1.0 + 1.5 + 0.5 + 2.0 + 1.0 = \text{Total} = 8 \text{ km}$$

Symbol

Location

H

home

R₁

Johat mor.

R₂

RJ mall

R₃

Rashid minhas Road

U

University Road

N

Nipa Choranghi

G

FUUAAT

K₁

Komran choranghi

K₂

Ruffi Mall

U

Uni Road

Dated: _____

N Nipa cherangi

G FUAAT

B1 Bela Jista

B2 Continental Bakery

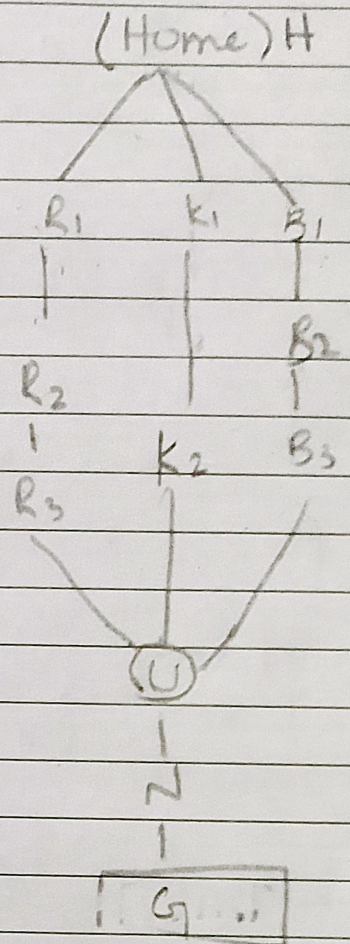
B3 Samanna mall

U Uri Road

N Nipa cherangi

G FUAAT

Tree Structure



Step: 0: Initialization

$Q: [H]$

$V: [H]$

Step: 01: Dequeue H Expand $H \rightarrow (R, K, B)$

$Q: [R, K, B]$

$V: [H, R, K, B]$

Step: 02: Dequeue R Expand $R \rightarrow (R_2)$

$Q: [K, B, R_2]$

$V: [H, R, K, B, R_2]$

Step: 03: Dequeue K Expand $K \rightarrow (K_2)$

$Q: [B, R_2, K_2]$

$V: [H, R, K, B, R_2, K_2]$

Step: 04: Dequeue B Expand $B \rightarrow (B_2)$

$Q: [R_2, K_2, B_2]$

$V: [H, R, K, B, R_2, K_2, B_2]$

Dated: _____

Step:05: Dequeue R_2 Expand $R_2 \rightarrow (R_3)$

$Q: [K_2, B_2, R_3]$

$V: [H, R_1, K_1, B_1, R_2, K_2, B_2, R_3]$

Step:06: Dequeue K_2 Expand $K_2 \rightarrow (U)$

$Q: [B_2, R_3, U]$

$V: [H, R_1, K_1, B_1, R_2, K_2, B_2, R_3, U]$

Step:07: Dequeue B_2 Expand $B_2 \rightarrow (B_3)$

$Q: [R_3, U, B_3]$

$V: [H, R_1, K_1, B_1, R_2, K_2, B_2, R_3, U, B_3]$

Step:08: Dequeue R_3 Expand $R_3 \rightarrow (U) \rightarrow$ Already visited.

$Q: [U, B_3]$

$V: [H, R_1, K_1, B_1, R_2, K_2, B_2, R_3, U, B_3]$

Step:09: Dequeue U Expand $U \rightarrow (Z)$

$Q: [B_3, Z]$

$V: [H, R_1, K_1, B_1, R_2, K_2, B_2, R_3, U, B_3, Z]$

Step:10: Dequeue B_3 Expand $B_3 \rightarrow (U) \rightarrow$ Already visited

$Q: [Z]$

$V: [H, R_1, K_1, B_1, R_2, K_2, B_2, R_3, U, B_3, Z]$

Step:11: Dequeue Z Expand $Z \rightarrow (G)$

$Q: [G]$

$V: [H, R_1, K_1, B_1, R_2, K_2, B_2, R_3, U, B_3, Z, G]$

Step:12: Dequeue G

Goal found

Dated: _____

BFS Order Of Expansion:

$H \rightarrow R_1 \rightarrow K_1 \rightarrow B_1 \rightarrow R_2 \rightarrow K_2 \rightarrow B_2 \rightarrow R_3 \rightarrow U \rightarrow B_3 \rightarrow N \rightarrow G$

Shortest Path:

$H \rightarrow K_1 \rightarrow K_2 \rightarrow U \rightarrow N \rightarrow G$